

CS302JSC — JavaScript

Practical Lab Session 4

Making Decisions in Code: if-else & switch-case

BCA / BIT — Semester 5 | Dev Sanskriti Vishwavidyalaya

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Semester: 5th

Part 0 — From Ternary to if-else

Task 0.1

```
// The OLD way (ternary) – from Lab 2
let marks = 68;
let result = marks >= 40 ? "Pass" : "Fail";
console.log(result);

// The NEW way (if-else) – today's topic
let marks2 = 68;
if (marks2 >= 40) {
  console.log("Pass");
} else {
  console.log("Fail");
}
```

Output:

Pass
Pass

Both versions print the same answer, "Pass", because they check the exact same condition (marks >= 40).

Which style is easier to read? The if-else version is easier for me to read because the words "if" and "else" clearly show the two possible paths, while the ternary version squeezes the same logic into one line that is harder to scan at a glance.

Part 1 — The if Statement

Task 1.1 — Pass check with two test values

```
let marks = 55;
if (marks >= 40) {
  console.log("You passed!");
}

let marks2 = 30;
if (marks2 >= 40) {
  console.log("You passed!");
}
console.log("Done checking marks2");
```

Output:
You passed!
Done checking marks2

With marks = 55, the condition is true, so "You passed!" is printed. With marks2 = 30, the condition is false, so nothing is printed for that check — the code inside the { } is simply skipped.

Task 1.2 (Fun Task) — Can You Watch This Movie?

```
function checkMovieAccess(age) {
  if (age >= 18) {
    console.log("You can watch this movie");
  }
  console.log("Checked age: " + age);
}

checkMovieAccess(8); // a child
checkMovieAccess(15); // a teenager
checkMovieAccess(25); // an adult
```

Output:
Checked age: 8
Checked age: 15
You can watch this movie
Checked age: 25

Only the adult (age 25) satisfies age >= 18, so only that call prints "You can watch this movie". The child and teenager calls skip that line entirely.

Part 2 — if-else: Two Paths

Task 2.1 — Even or Odd (3 numbers)

```
function checkEvenOdd(number) {  
  if (number % 2 === 0) {  
    console.log(number + " is Even number");  
  } else {  
    console.log(number + " is Odd number");  
  }  
}  
  
checkEvenOdd(7);  
checkEvenOdd(12);  
checkEvenOdd(0);
```

Output:

```
7 is Odd number  
12 is Even number  
0 is Even number
```

Task 2.2 (Fun Task) — ATM PIN Checker

```
let correctPIN = 1234;  
  
let guess1 = 1234; // correct PIN  
if (guess1 === correctPIN) {  
  console.log("Access Granted");  
} else {  
  console.log("Access Denied");  
}  
  
let guess2 = 4321; // wrong PIN  
if (guess2 === correctPIN) {  
  console.log("Access Granted");  
} else {  
  console.log("Access Denied");  
}
```

Output:

```
Access Granted  
Access Denied
```

Task 2.3 — Compare with Lab 2 (Pass/Fail)

```
// Lab 2 version (ternary)
// let result = marks >= 40 ? "Pass" : "Fail";

// Rewritten with if-else
let marks = 45;
let result;
if (marks >= 40) {
  result = "Pass";
} else {
  result = "Fail";
}
console.log(result);
```

Output:
Pass

Which do I like better? For a simple pass/fail check the ternary is shorter and fine, but I prefer if-else overall because it is easier to extend later (e.g. adding more conditions or running multiple lines of code).

Part 3 — if-else-if: More Than Two Paths

Task 3.1 (Fun Task) — Movie Ticket Price (5 ages)

```
function ticketPrice(age) {
  let price;
  if (age < 5) {
    price = "Free";
  } else if (age < 12) {
    price = "Rs. 100";
  } else if (age < 60) {
    price = "Rs. 250";
  } else {
    price = "Rs. 150";
  }
  console.log("Age " + age + " -> " + price);
}
ticketPrice(3);
ticketPrice(9);
ticketPrice(30);
ticketPrice(65);
ticketPrice(12);
```

Output:

```
Age 3 -> Free
Age 9 -> Rs. 100
Age 30 -> Rs. 250
Age 65 -> Rs. 150
Age 12 -> Rs. 250
```

Note that age 12 falls into the "age < 60" bracket (Rs. 250), not the "age < 12" bracket, because JavaScript checks conditions top to bottom and 12 < 12 is false.

Task 3.2 (Fun Task) — Weather Advice Bot (4 temperatures)

```
function weatherAdvice(temp) {
  let advice;
  if (temp > 35) {
    advice = "It's hot! Drink water.";
  } else if (temp > 20) {
    advice = "Nice weather!";
  } else if (temp > 10) {
    advice = "A bit cold. Wear a jacket.";
  } else {
    advice = "Very cold! Stay warm.";
  }
  console.log(temp + "°C -> " + advice);
}

weatherAdvice(40);
weatherAdvice(25);
weatherAdvice(15);
weatherAdvice(2);
```

Output:

```
40°C -> It's hot! Drink water.
25°C -> Nice weather!
15°C -> A bit cold. Wear a jacket.
2°C -> Very cold! Stay warm.
```

Part 4 — switch-case: A Clean Way for Many Choices

Task 4.1 — Days of the Week (1–7)

```
function printDay(day) {
  switch (day) {
    case 1:
      console.log("Monday");
      break;
```

```
case 2:
  console.log("Tuesday");
  break;
case 3:
  console.log("Wednesday");
  break;
case 4:
  console.log("Thursday");
  break;
case 5:
  console.log("Friday");
  break;
case 6:
  console.log("Saturday");
  break;
case 7:
  console.log("Sunday");
  break;
default:
  console.log("Invalid day number");
}
}

printDay(3);
printDay(7);
printDay(10);
```

```
Output:
Wednesday
Sunday
Invalid day number
```

Task 4.2 (Fun Task) — Mood Emoji Switch

```
function moodMessage(mood) {  
  switch (mood) {  
    case "happy":  
      console.log("💎 Keep smiling!");  
      break;  
    case "sad":  
      console.log("💎 It's okay, things will get better.");  
      break;  
    case "angry":  
      console.log("💎 Take a deep breath.");  
      break;  
    case "tired":  
      console.log("💎 Time to rest a bit!");  
      break;  
    default:  
      console.log("💎 Thanks for sharing how you feel!");  
  }  
}  
  
moodMessage("happy");  
moodMessage("angry");  
moodMessage("excited");
```

Output:

```
💎 Keep smiling!  
💎 Take a deep breath.  
💎 Thanks for sharing how you feel!
```

Task 4.3 — The Missing break Experiment

Original Task 4.1 code (with all break statements) called with printDay(3) correctly prints only "Wednesday".

Now the break is removed from case 3 on purpose:

```
case 3:  
  console.log("Wednesday");  
  // break; <-- break removed on purpose  
case 4:  
  console.log("Thursday");  
  break;
```

Calling printDay(3) with this broken version:

```
Output:  
Wednesday  
Thursday
```

Even though day is 3, both "Wednesday" AND "Thursday" get printed. This happens because without break, JavaScript doesn't stop after a matching case — it keeps running the code in the next case too, until it finds a break or reaches the end of the switch. This is called "fall-through". After putting the break back, only "Wednesday" prints again.

In my own words: break tells the switch statement to stop checking any further cases once a match has been found and its code has run. Without break, JavaScript falls through and keeps executing the code of the following cases too, even though their condition wasn't matched.

Part 5 — Mini Project: Simple ATM Machine

A real ATM first checks the PIN (if-else), and only if it is correct does it show a menu of choices (switch case). The withdraw option additionally uses an if-else inside its case to check for insufficient funds.

```
let correctPIN = 1234;
let enteredPIN = 1234; // change this to test a wrong PIN
let balance = 5000;

if (enteredPIN === correctPIN) {
  // PIN is correct — show the menu
  let choice = 3; // 1 = Check Balance, 2 = Withdraw, 3 = Deposit

  switch (choice) {
    case 1:
      console.log("Your current balance is: Rs. " + balance);
      break;

    case 2: {
      let withdrawAmount = 7000;
      if (withdrawAmount > balance) {
        console.log("Insufficient funds");
      } else {
        balance = balance - withdrawAmount;
        console.log("Withdrawal successful. New balance: Rs. " + balance);
      }
      break;
    }

    case 3: {
      let depositAmount = 2000;
      balance = balance + depositAmount;
      console.log("Deposit successful. New balance: Rs. " + balance);
      break;
    }

    default:
      console.log("Invalid choice");
  }
} else {
  console.log("Wrong PIN. Access Denied.");
}
```

Test (a) — Wrong PIN

enteredPIN = 9999 (correctPIN stays 1234):

Output:
Wrong PIN. Access Denied.

Test (b) — Correct PIN, withdrawal bigger than balance

enteredPIN = 1234, choice = 2, withdrawAmount = 7000, balance = 5000:

Output:
Insufficient funds

Test (c) — Correct PIN, successful deposit

enteredPIN = 1234, choice = 3, depositAmount = 2000, balance = 3000:

Output:
Deposit successful. New balance: Rs. 5000

These results match the sanity-check numbers given in the lab sheet: wrong PIN → "Wrong PIN. Access Denied."; withdrawing 7000 from a balance of 3000 → "Insufficient funds"; depositing 2000 into 3000 → new balance 5000.

Part 6 — Debugging Challenge

Snippet 1

Buggy code:

```
let age = 15;
if (age = 18) { // BUG: single = is assignment, not comparison
  console.log("Adult");
} else {
  console.log("Minor");
}
```

What was wrong: The condition uses a single "=" which is the assignment operator, not the comparison operator. This assigns 18 to age (so age becomes 18) and, since 18 is a truthy value, the if block always runs — printing "Adult" no matter what age started as.

Fixed code:

```
let age = 15;
if (age === 18) { // FIX: use === for comparison
  console.log("Adult");
} else {
  console.log("Minor");
}
```

Output:
Minor

Snippet 2

Buggy code:

```
let fruit = "apple";
switch (fruit) {
  case "apple":
    console.log("Red fruit");
    // BUG: missing break here causes fall-through
  case "banana":
    console.log("Yellow fruit");
    break;
  default:
    console.log("Unknown fruit");
}
```

What was wrong: There is no break after the "apple" case, so after printing "Red fruit" the code falls through into the "banana" case and also prints "Yellow fruit", even though fruit is "apple", not "banana".

Fixed code:

```
let fruit = "apple";
switch (fruit) {
  case "apple":
    console.log("Red fruit");
    break; // FIX: added break
  case "banana":
    console.log("Yellow fruit");
    break;
  default:
    console.log("Unknown fruit");
}
```

Output:
Red fruit

Snippet 3

Buggy code:

```
let choice = "2"; // BUG: this is a string "2", not the number 2
switch (choice) {
  case 1:
    console.log("One");
    break;
  case 2:
    console.log("Two");
    break;
  default:
    console.log("Invalid");
}
```

What was wrong: switch uses strict comparison (===) internally, so it checks type as well as value. choice is the string "2", but the cases compare against the number 2. Since "2" === 2 is false, none of the cases match and the default runs, printing "Invalid" even though the value looks correct.

Fixed code (option 1 — fix the variable's type):

```
let choice = 2; // FIX: use a number instead of a string
switch (choice) {
  case 1:
    console.log("One");
    break;
  case 2:
    console.log("Two");
    break;
  default:
    console.log("Invalid");
}
```

Output:
Two

Fixed code (option 2 — convert before switching, if choice must stay a string):

```
let choice = "2";
switch (Number(choice)) { // FIX: convert string to number
  case 1:
    console.log("One");
    break;
  case 2:
    console.log("Two");
    break;
  default:
    console.log("Invalid");
}
```

Output:

Two